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Machine Learning Assignment Report: Titanic Survival Prediction

Subject: Machine Learning

Trimester: Trimester 2

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Assignment objective

Build and compare a complete supervised learning pipeline for Titanic survival prediction, covering preprocessing, baseline models, decision trees, rule extraction, kNN, and ensemble methods.

Report basis

Dataset: Titanic (891 passengers, 12 columns).
Results validated from executed notebook (43 cells, 0 errors, 15 figures).

1. Dataset and Preprocessing

The Titanic dataset (Kaggle) contains information on 891 passengers with 12 features. The binary target Survived (0/1) is moderately imbalanced — 549 non-survivors vs 342 survivors — so evaluation emphasized F1-score alongside accuracy.

Property	Value
Rows	891
Columns	12
Target column	Survived
Class balance	Not Survived: 549 Survived: 342
Missing values	Age: 177 (19.9%) Cabin: 687 (77.1%) Embarked: 2
Duplicate rows	0
Train / Test split	712 / 179 (80% / 20%, stratified by target)

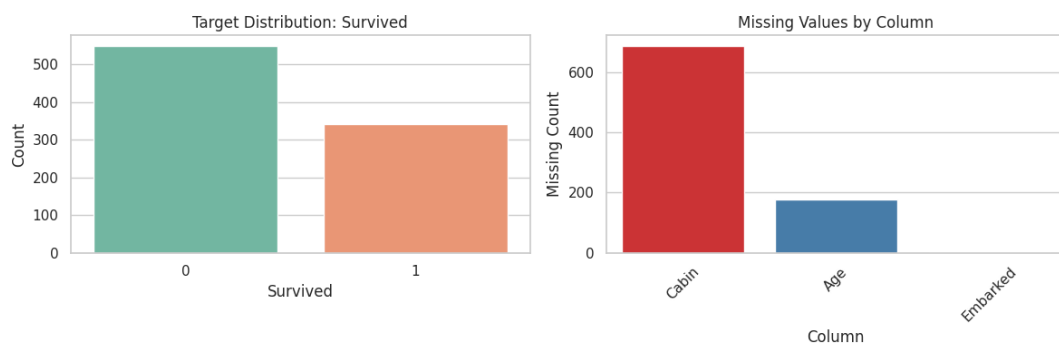


Figure 1. Target class distribution (left) and missing-value counts by column (right). Cabin is too sparse to impute reliably.

Attribute type	Columns	Handling
Ordinal	Pclass	Used as-is (numeric: 1=First class, 3=Third class)
Nominal	Sex, Embarked	One-hot encoded: Sex_female/male, Embarked_C/Q/S
Numeric cont.	Age, Fare	Median imputation; StandardScaler applied for LR / NB / kNN
Numeric disc.	SibSp, Parch	Used as-is; IQR outlier analysis performed
Excluded	PassengerId, Name, Ticket, Cabin	Identifier-like, high-cardinality, or >77% missing — dropped

Preprocessing rationale

PassengerId, Name, Ticket and Cabin were dropped — they are identifiers or mostly missing. Age was imputed with the median (robust to its right skew). Embarked used mode imputation (only 2 missing). Fare outliers were retained because very high fares correspond to genuine first-class passengers, not data errors. Two preprocessor pipelines were built: one with StandardScaler (for LR, NB, kNN) and one without scaling (for all tree-based models).

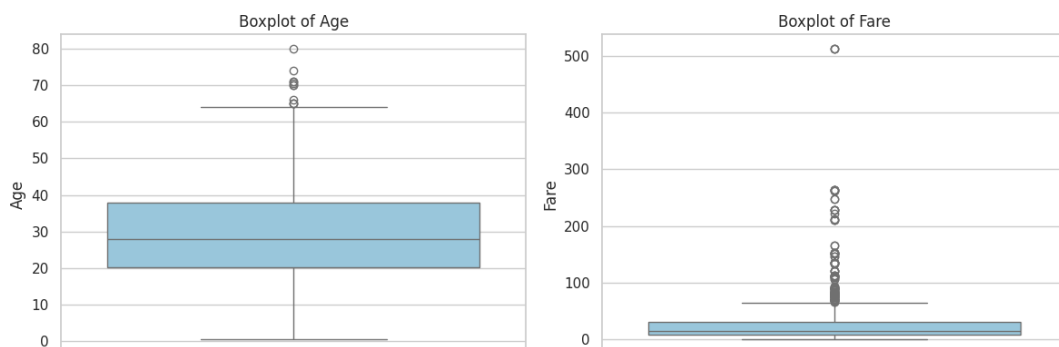


Figure 2. Age and Fare boxplots confirm right-skewed distributions and high-value outliers. Outliers were kept — they represent real fare differences between passenger classes.

2. Baseline Models and Decision Tree

Baseline model results

Logistic Regression and Gaussian Naive Bayes were trained as baselines on the scaled feature set (StandardScaler + one-hot encoding, 80/20 stratified split).

Model	Accuracy	F1	ROC-AUC
Logistic Regression	0.8045	0.7244	0.8437
Gaussian Naive Bayes	0.7877	0.7164	0.8191

Logistic Regression (accuracy 0.8045, F1 0.7244, ROC-AUC 0.8437) outperformed Naive Bayes across all metrics. Both establish a reference floor — any more complex model that cannot beat these numbers is not adding useful predictive value.

Decision tree training and tuning

An untuned Decision Tree was trained first (accuracy 0.8268, F1 0.7634). 5-fold GridSearchCV then searched $\text{max_depth} \in \{3,4,5,6,\text{None}\}$, $\text{min_samples_split} \in \{2,5,10,20\}$, $\text{min_samples_leaf} \in \{1,2,4,6\}$. Best CV params: $\text{max_depth}=\text{None}$, $\text{min_samples_split}=10$, $\text{min_samples_leaf}=1$ (CV F1=0.7502). The tuned tree scored 0.7111 F1 on the test set — worse than the untuned tree. CV ranking and held-out ranking can diverge, especially with small datasets.

Feature importance and decision paths

Sex_female is the dominant predictor, followed by Age, Fare, and Pclass. Embarkation port contributes very little. Two representative paths from the untuned tree:

- **Path 1 (Survived):** Female, Pclass ≤ 2.5 , Fare > 28.86 , Parch ≤ 1.5 — 71 training passengers, purity 1.00.
- **Path 2 (Did Not Survive):** Male, Age > 3.5 , Pclass > 1.5 , Fare ≤ 7.13 — large group, purity 0.90+.

Feature	Gini Importance
Sex_female	0.3150
Age	0.2698
Fare	0.2349
Pclass	0.1097
Embarked_S	0.0227
SibSp	0.0195
Parch	0.0121

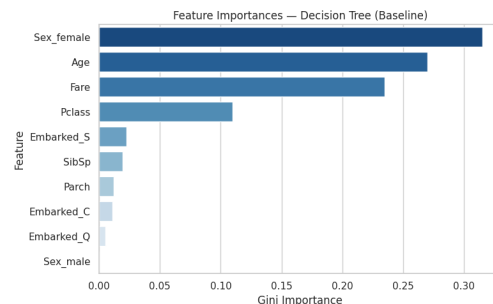


Figure 3 (left). Feature importance scores. Figure 3 (right). Horizontal bar confirms Sex_female, Age, and Fare dominate — Embarked_S and SibSp/Parch contribute very little.

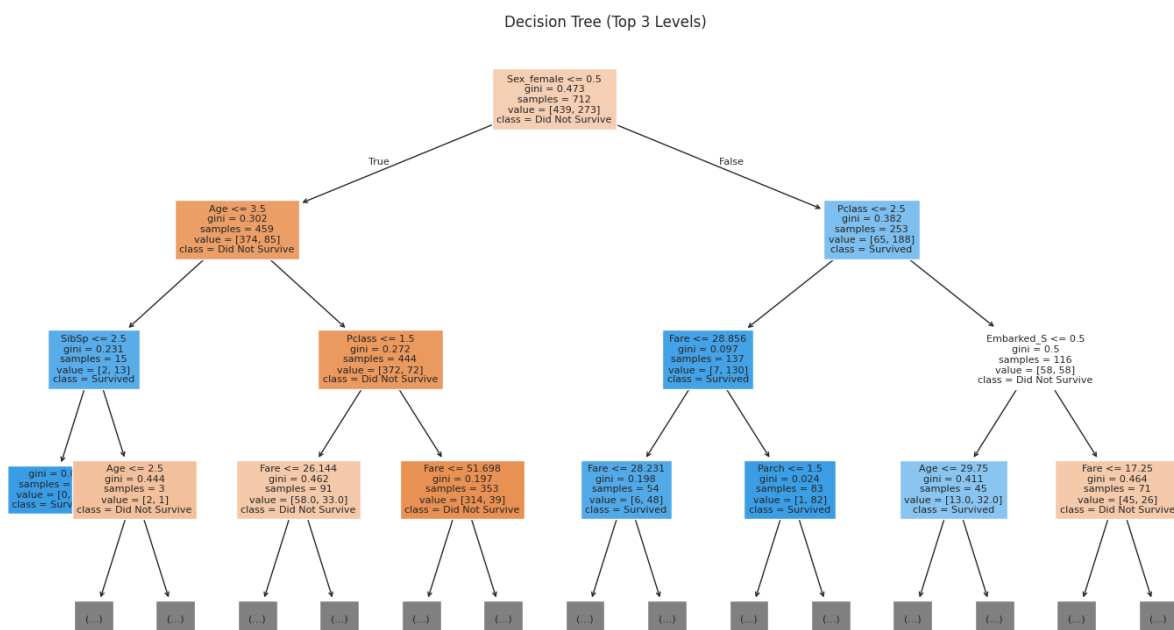


Figure 4. Decision tree top 3 levels. First split on Sex_female; subsequent splits on Fare, Age, and Pclass reflect known survival patterns from the historical record.

3. Rule-Based Classification and kNN

Rule-based classification

A constrained decision tree (max_depth=4, min_samples_split=20, min_samples_leaf=10) was trained to generate short, human-readable rules. It achieved accuracy 0.7765 and F1 0.6552 — lower than the full tree, but every decision is expressible as a plain if-then statement with measurable support and purity.

Rule	Prediction	n	Purity
1. IF Male, age > 3.5, Pclass > 1.5, fare ≤ 51.70	Did Not Survive	339	0.900
2. IF Female, Pclass ≤ 2.5, fare > 28.86, Parch ≤ 1.5	Survived	71	1.000
3. IF Female, Pclass ≤ 2.5, fare ≤ 28.86, age ≤ 37	Survived	36	0.944
4. IF Female, Pclass > 2.5, Embarked=S, fare > 17.25	Did Not Survive	27	0.852
5. IF Female, Pclass > 2.5, Embarked≠S, fare ≤ 8.08	Survived	23	0.870

Table 3. Five rules extracted from the constrained tree with support (n = training passengers) and purity. Rule 2 (purity 1.00, 71 passengers) is perfectly reliable.

Interpretability trade-off: the rule-based model is easy to audit and communicate — a stakeholder can verify each rule without ML knowledge. The cost is a drop in recall (0.55 vs 0.72 for the full tree), meaning the model misses more survivors.

kNN (Lazy Learning)

kNN was evaluated with $k \in \{1, 3, 5, 7, 9\}$ and two distance metrics (Euclidean, Manhattan). All experiments used 5-fold cross-validation on the training set with preprocessing inside each fold to avoid data leakage. Best result: Manhattan distance, $k=9$ (CV F1=0.7574, test accuracy=0.7989, test F1=0.7188, ROC-AUC=0.8432).

Why scaling is essential for kNN: Fare ranges up to £512 while SibSp ranges 0–8. Without StandardScaler, distance is dominated entirely by Fare, making the other features irrelevant. After scaling, all features contribute proportionally.

Bias-variance trade-off: $k=1$ memorises training data (low bias, high variance — overfits). As k increases, the boundary smooths and variance falls but bias rises. Performance peaked at $k=7-9$, indicating the data contains enough noise that very local neighbourhoods hurt generalisation.

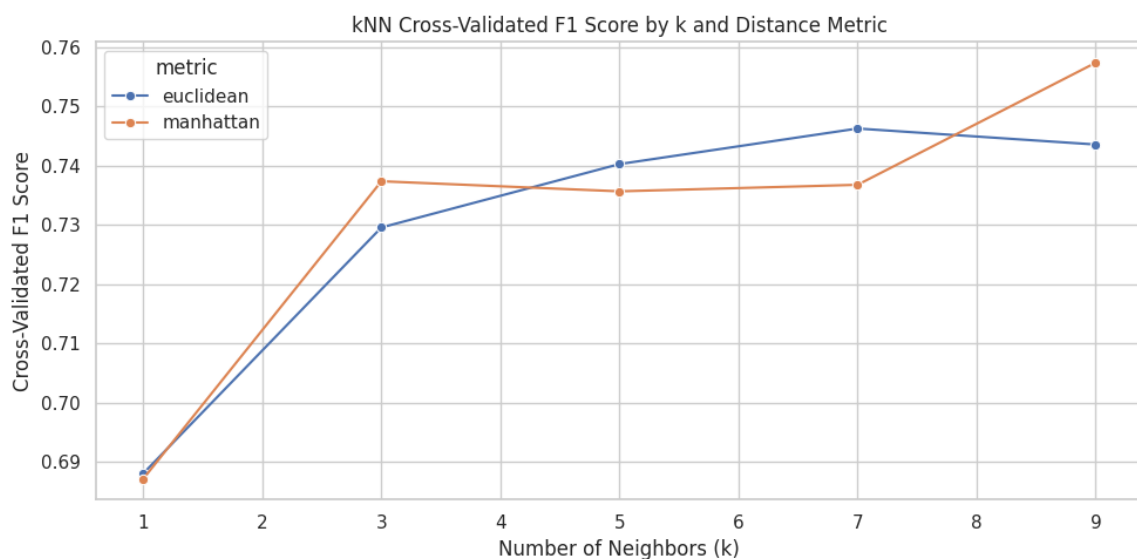


Figure 5. Cross-validated F1 score vs k for Euclidean and Manhattan distance. Manhattan consistently outperforms Euclidean; both peak near $k=7-9$.

4. Ensemble Learning and Final Comparison

Ensemble results

Three ensemble methods were trained: Random Forest (300 trees, bagging), Gradient Boosting (sequential boosting), and AdaBoost (adaptive boosting). All used the unscaled tree preprocessor (imputation + one-hot encoding only).

Surprisingly, none of the ensemble models beat the plain untuned Decision Tree. Random Forest matched Logistic Regression on accuracy (0.8045) but trailed on F1 (0.7154 vs 0.7634). Gradient Boosting and AdaBoost were weaker still. This is expected for small, low-dimensional datasets where the dominant features (sex, age, fare) are already captured by a single well-structured tree.

Final model comparison — all 9 models

Model	Acc	Prec	Rec	F1	AUC
Decision Tree (Baseline)	0.827	0.806	0.725	0.763	0.803
Logistic Regression	0.804	0.793	0.667	0.724	0.844
kNN (k=9, manhattan)	0.799	0.780	0.667	0.719	0.843
Gaussian Naive Bayes	0.788	0.739	0.696	0.716	0.819
Random Forest	0.804	0.815	0.638	0.715	0.840
Gradient Boosting	0.799	0.789	0.652	0.714	0.818
Decision Tree (Tuned)	0.782	0.727	0.696	0.711	0.800
AdaBoost	0.782	0.750	0.652	0.698	0.825
Rule-Based Tree	0.776	0.808	0.551	0.655	0.814

Table 4. All models ranked by F1 score (descending). Highlighted row = best overall. Acc = Accuracy, Prec = Precision, Rec = Recall, AUC = ROC-AUC.

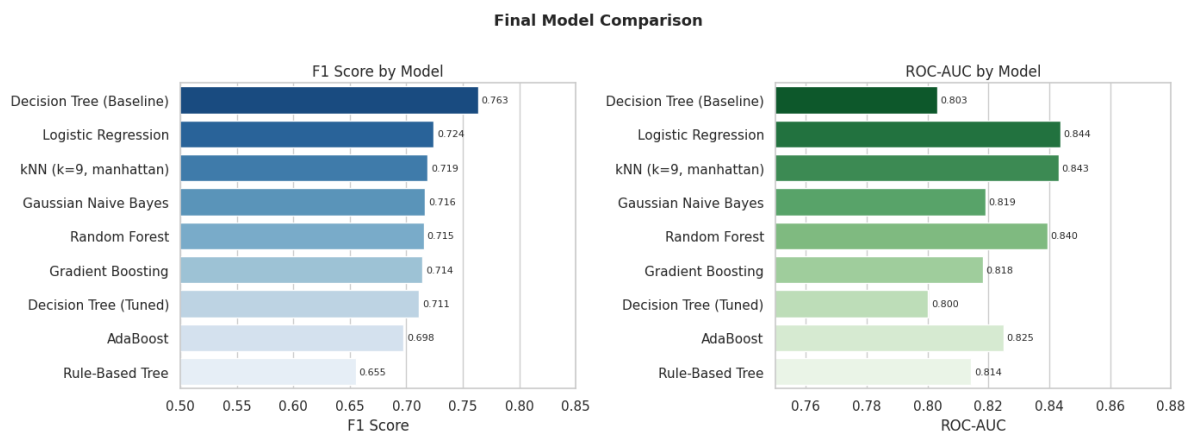


Figure 6. F1 score (left) and ROC-AUC (right) for all 9 models. Decision Tree (Baseline) leads on F1; Logistic Regression leads on ROC-AUC.

Key insights

- Decision Tree (Baseline): best F1 (0.763) and accuracy (0.827) — top overall model.
- Logistic Regression: best ROC-AUC (0.844) — most reliable probability ranking.
- kNN: competitive only after scaling; confirms preprocessing importance.
- Rule-Based Tree: lowest F1 but most interpretable — best for explanation-first use.
- Ensembles: did not gain over simpler models here — complexity $\neq$ guaranteed improvement.

5. Conclusions, Risks, and Recommendations

Best-performing model

The untuned Decision Tree achieved the highest F1 (0.7634) and accuracy (0.8268) on the held-out test set. It captures non-linear interactions between sex, class, age, and fare that a linear model cannot represent, without requiring the additional complexity of ensemble aggregation. Interestingly, the GridSearchCV-tuned variant scored worse on the test set (F1 0.7111), showing that CV performance does not always translate to held-out performance on small datasets.

Most interpretable model

The Rule-Based Tree is the most interpretable. Its five extracted rules can be written as plain if-then statements that a non-technical audience can read and verify without understanding machine learning. Each rule includes the number of training passengers it covers and its purity, giving statistical backing to the plain-language explanation. The cost is a drop in recall from 0.72 to 0.55.

Deployment risks

- Historical bias: the model is trained on 1912 data — survival patterns reflect social norms of that era and cannot generalize to modern contexts.
- Fairness: sex and passenger class are the strongest predictors, so the model makes systematically different predictions for men vs women and rich vs poor passengers.
- Threshold brittleness: tree splits create hard boundaries (e.g. $\text{Fare} \leq 7.13 \rightarrow \text{non-survival}$). A passenger just above or below that threshold gets opposite predictions.
- Missing data sensitivity: Age is missing for ~20% of passengers; the imputation strategy directly affects predictions for those passengers in deployment.
- Small dataset: 891 rows is insufficient for stable deployment or strong confidence intervals on out-of-distribution inputs.

Limitations of this analysis

- Single train-test split (`random_state=42`) — results may vary with different seeds.
- No repeated cross-validation or bootstrap resampling for confidence intervals.
- Feature engineering (e.g. family size, title from Name) was intentionally excluded to keep the pipeline transparent and aligned with the assignment scope.
- XGBoost was not tested (not available in the environment); Gradient Boosting from sklearn was used as the boosting representative.

Recommendations

For maximum predictive performance on this dataset: use the baseline Decision Tree. For stakeholder communication and explainability: use the Rule-Based Tree. For a compact statistical baseline: Logistic Regression offers stable ROC-AUC (0.844) with a simple, well-understood model form. Future work could improve robustness through repeated cross-validation, threshold tuning, and title-based feature engineering.

Conclusion

This assignment built a complete supervised learning pipeline from raw Titanic data to a final model comparison across nine classifiers. Careful preprocessing, stratified evaluation, and explicit interpretability analysis showed that a simple Decision Tree can outperform ensemble methods when features are low-dimensional and highly predictive. The most important result is not which model won numerically, but that the analysis exposes a concrete trade-off: the Rule-Based Tree loses 10 F1 points relative to the best tree, but gains full human-readable explainability — a trade-off that matters enormously in any real deployment context.